Basic chemistry for engineers

Engineers meet chemistry everywhere: in concrete that sets, steel that rusts, in batteries, fuels and waste water. DTH's basic chemistry course was written for the three large lines – civil (B), electrical (E) and mechanical (M) engineering – and gave the foundation: how substances react, how far and how fast. The page has an interactive titration curve and an Arrhenius plot.

> DTH, course 2010 (1986), 6 points: Basic chemistry (for the B, E and M lines). Chemistry later became one of my subjects as a teacher at the gymnasium (upper secondary school).

Atoms and bonds

Chemistry is the physics of electrons. The periodic table orders the elements by nuclear charge (Moseley 1913) and reflects the shell structure of the electrons, which quantum mechanics explains (see atomic physics). Atoms bond to achieve a stable electron configuration: in *ionic bonds* electrons are transferred (NaCl), in *covalent bonds* they are shared (H_2O, CH_4), and in *metallic bonds* they are shared by the whole crystal, giving metals their conductivity and ductility. Linus Pauling's electronegativity scale (1932) tells how polar a bond is. Between molecules weaker forces act – van der Waals forces and hydrogen bonds – that set boiling points and solubility (see properties of matter).

Stoichiometry

A reaction equation is a balance sheet: atoms are conserved. Amount of substance is measured in moles, $n = m/M$, where one mole contains $N_A = 6.022 \cdot 10^{23}$ particles. The combustion of propane in a gas boiler,

$$\mathrm{C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O}$$

shows that 44 g of propane needs 160 g of oxygen and gives 132 g of CO_2. For gases the ideal gas law $pV = nRT$ holds, so one mole occupies 24.5 L at 25 °C and 1 atm. The engineer uses stoichiometry to calculate the air requirement of a boiler, the amount of lime to neutralise waste water or the CO_2 emission of a cement works.

Thermochemistry

The enthalpy of reaction ΔH is the heat given off or taken up at constant pressure. Germain Hess found in 1840 that it depends only on the initial and final states, not on the path – so it can be calculated from tabulated enthalpies of formation: $\Delta H^\circ = \sum \Delta H_f^\circ(\text{products}) - \sum \Delta H_f^\circ(\text{reactants})$. Whether a reaction proceeds by itself is, however, decided not by ΔH alone but by the Gibbs free energy

$$\Delta G = \Delta H - T\Delta S, \qquad \Delta G^\circ = -RT\ln K$$

A reaction can be endothermic and still proceed if the entropy increases enough – as when salt dissolves in water and the water becomes cold. (More on entropy: thermodynamics and the arrow of time.)

Equilibrium

Most reactions do not go to completion but settle into an equilibrium where the forward and reverse reactions run equally fast. Guldberg and Waage – a Norwegian mathematician and his brother-in-law, a chemist – formulated the law of mass action in 1864: for $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$,

$$K = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$$

is constant at a given temperature. Le Chatelier's principle tells how the equilibrium shifts when pressure, temperature or concentrations are changed – the Haber–Bosch process for ammonia is the classic example (see Le Chatelier's principle).

Acids and bases

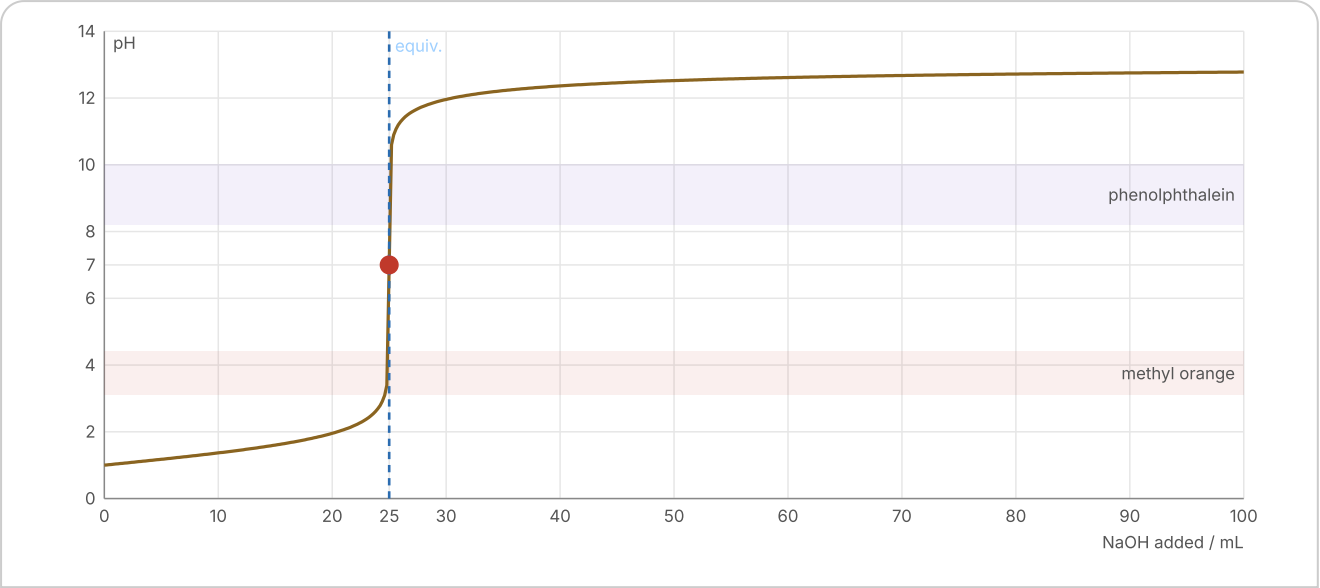
Johannes Nicolaus Brønsted of the University of Copenhagen and Thomas Lowry independently defined in 1923 an acid as a substance that gives up a proton and a base as one that accepts one. Water is both acid and base: $[\mathrm{H_3O^+}][\mathrm{OH^-}] = 10^{-14}$ at 25 °C. The pH scale, $\mathrm{pH} = -\log_{10}[\mathrm{H_3O^+}]$, was introduced in 1909 by S. P. L. Sørensen at the Carlsberg Laboratory. A weak acid is only partly protolysed, and its strength is given by $\mathrm{p}K_a$. A mixture of a weak acid and its conjugate base is a *buffer* that keeps the pH almost constant:

$$\mathrm{pH} = \mathrm{p}K_a + \log_{10}\frac{[\mathrm{A^-}]}{[\mathrm{HA}]} \qquad \text{(Henderson–Hasselbalch)}$$

Hasselbalch was Danish too. The blood is held at pH 7.4 by the carbonic acid–hydrogen carbonate buffer; concrete has a pH of about 13, which is what protects the reinforcing steel from rust – until the CO_2 of the air has neutralised the surface of the concrete (carbonation).

Titration

In a titration a base of known concentration is added to an acid while the pH is followed. The equivalence point, where as many moles of base as acid have been added, shows as a jump in pH. Half-way to the equivalence point $[\mathrm{HA}] = [\mathrm{A^-}]$, and then $\mathrm{pH} = \mathrm{p}K_a$ – that is how acid strength is measured.



Electrochemistry and corrosion

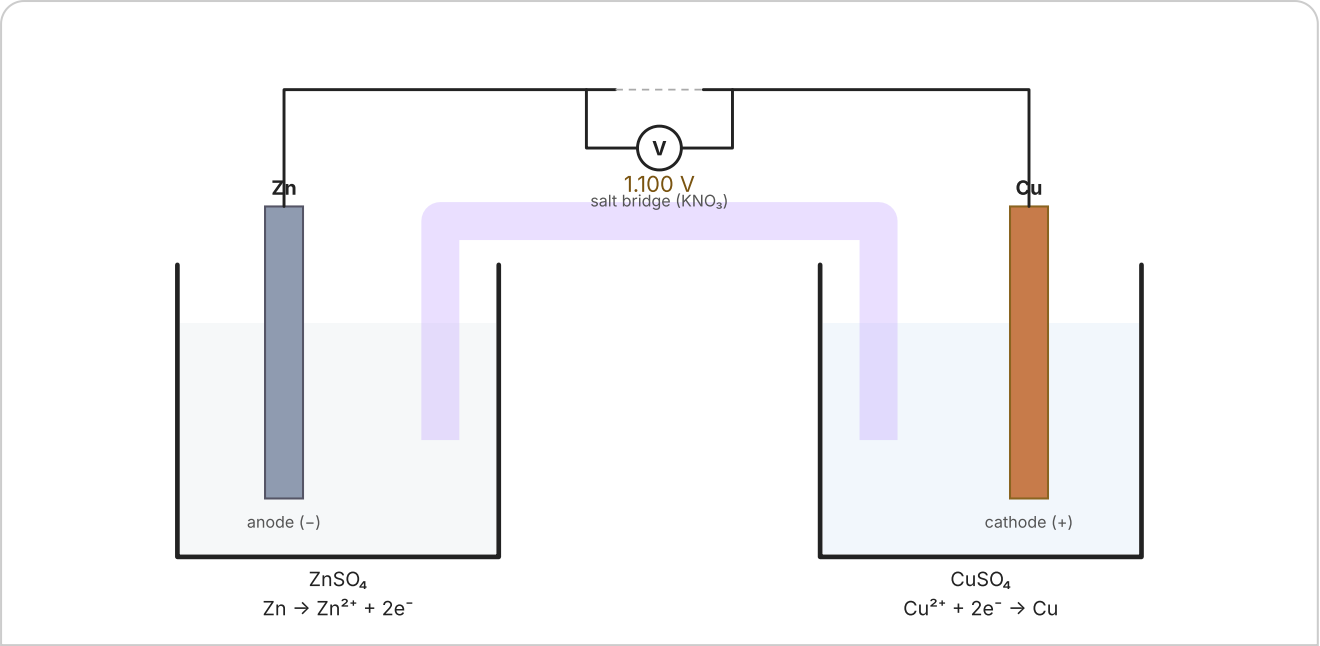
In a galvanic cell oxidation and reduction take place in separate places, and the electrons go through an external wire. In the Daniell cell zinc dissolves and copper is deposited; the cell voltage is 1.10 V under standard conditions. The voltage depends on the concentrations through Walther Nernst's equation (1889):

$$E = E^\circ - \frac{RT}{zF}\ln Q$$

where Q is the reaction quotient, z the number of electrons and F Faraday's constant. At 25 °C $RT/F \cdot \ln 10 = 59$ mV – a factor of ten in concentration moves a potential by 59 mV/z. That is the basis of the pH electrode.

Corrosion is electrochemistry one does not want. When two metals with different electrode potentials are in contact in a moist environment, the less noble metal becomes the anode and dissolves: a steel screw in a copper sheet rusts quickly. Even on a single piece of iron local anodes and cathodes can form. The engineer protects with paint, galvanising (zinc sacrifices itself for the

iron), sacrificial anodes on ships and harbour structures, or impressed current (cathodic protection) on pipelines.

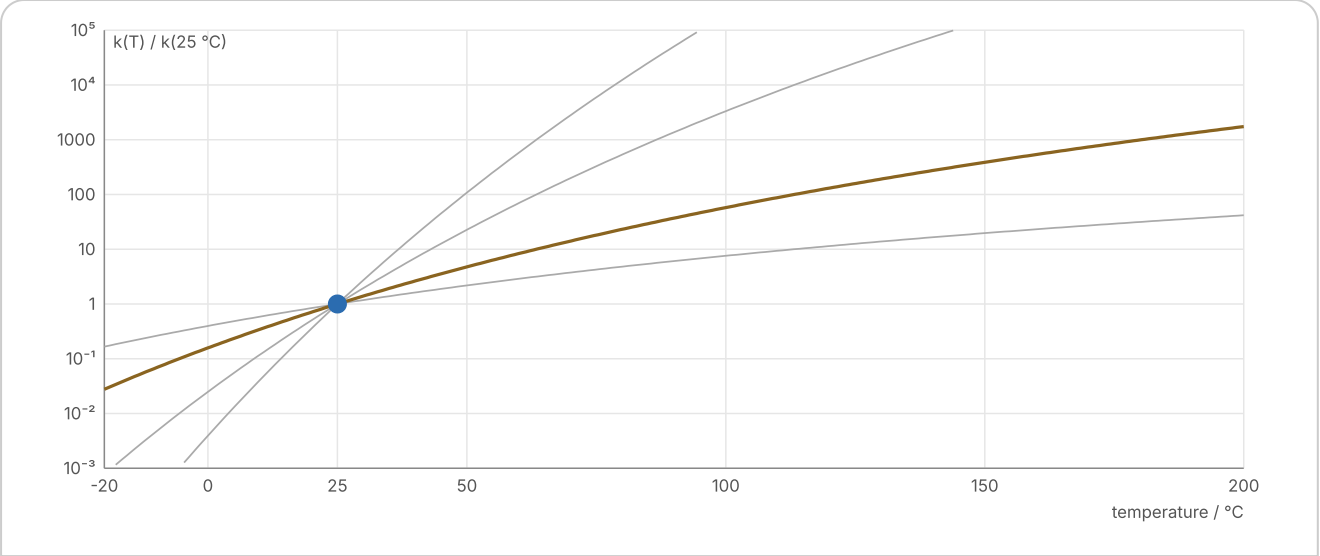


Reaction rate

Thermodynamics says whether a reaction can happen; kinetics, how fast. Svante Arrhenius found in 1889 that the rate constant grows exponentially with temperature, because the molecules must cross an energy barrier, the activation energy E_a:

$$k = A\,e^{-E_a/RT} \qquad \Longleftrightarrow \qquad \ln k = \ln A - \frac{E_a}{R}\cdot\frac{1}{T}$$

Plotting $\ln k$ against $1/T$ gives a straight line with slope $-E_a/R$. The old rule of thumb that the reaction rate doubles for every 10 °C corresponds to $E_a \approx 50$ kJ/mol at room temperature. A catalyst lowers the barrier without being used up itself – the car's three-way catalyst, enzymes and the iron catalyst of the Haber–Bosch process.



Since 1989

- **Green chemistry:** The focus has moved from yield to resources and waste: catalysis, non-toxic solvents and “atom economy”.
- **The lithium-ion battery** came on the market in 1991 (Sony); Goodenough, Whittingham and Yoshino received the Nobel Prize in 2019. Electrochemistry is now at the heart of the green transition – batteries, fuel cells and electrolysis for hydrogen and Power-to-X.
- **CO_2 and cement:** Cement production accounts for about 8 % of the world’s CO_2 emissions, half of it from the calcination itself, $\mathrm{CaCO_3 \rightarrow CaO + CO_2}$; new binders and CO_2 capture are in focus, including at Aalborg Portland.
- **Computational chemistry:** Density functional theory (Kohn and Pople, Nobel Prize 1998) computes reaction barriers and designs catalysts, among other places at DTU.

Further reading

- Peter Atkins, Loretta Jones and Leroy Laverman: *Chemical Principles* (7th ed., Freeman 2016)
- Theodore L. Brown et al.: *Chemistry: The Central Science* (15th ed., Pearson 2022)
- On this site: Le Chatelier’s principle · Chemistry self-test (in Danish)